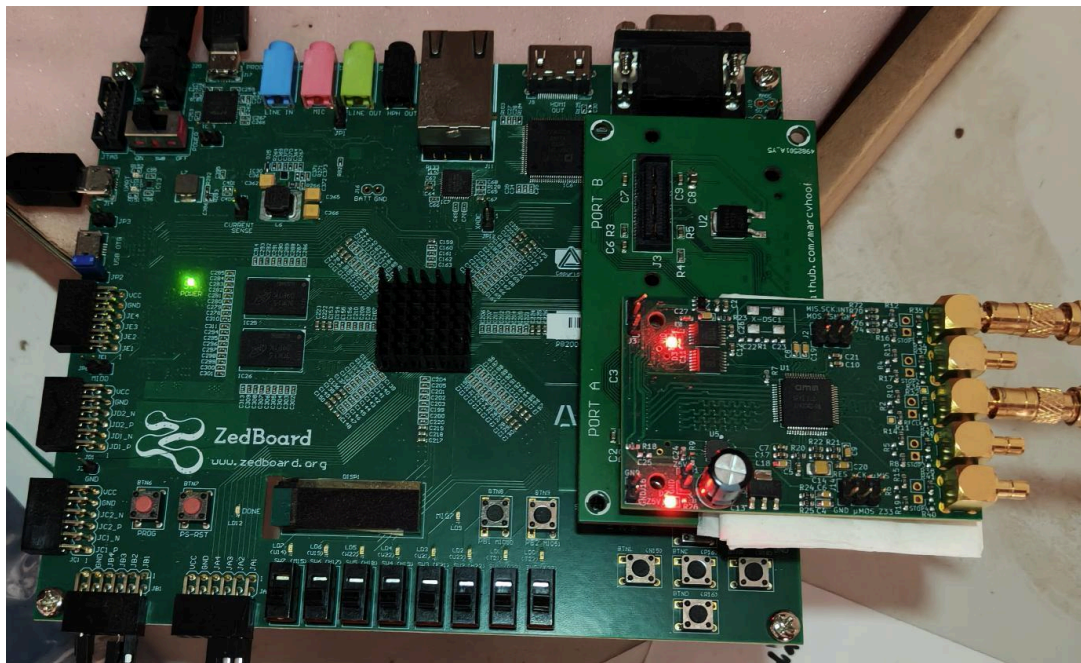


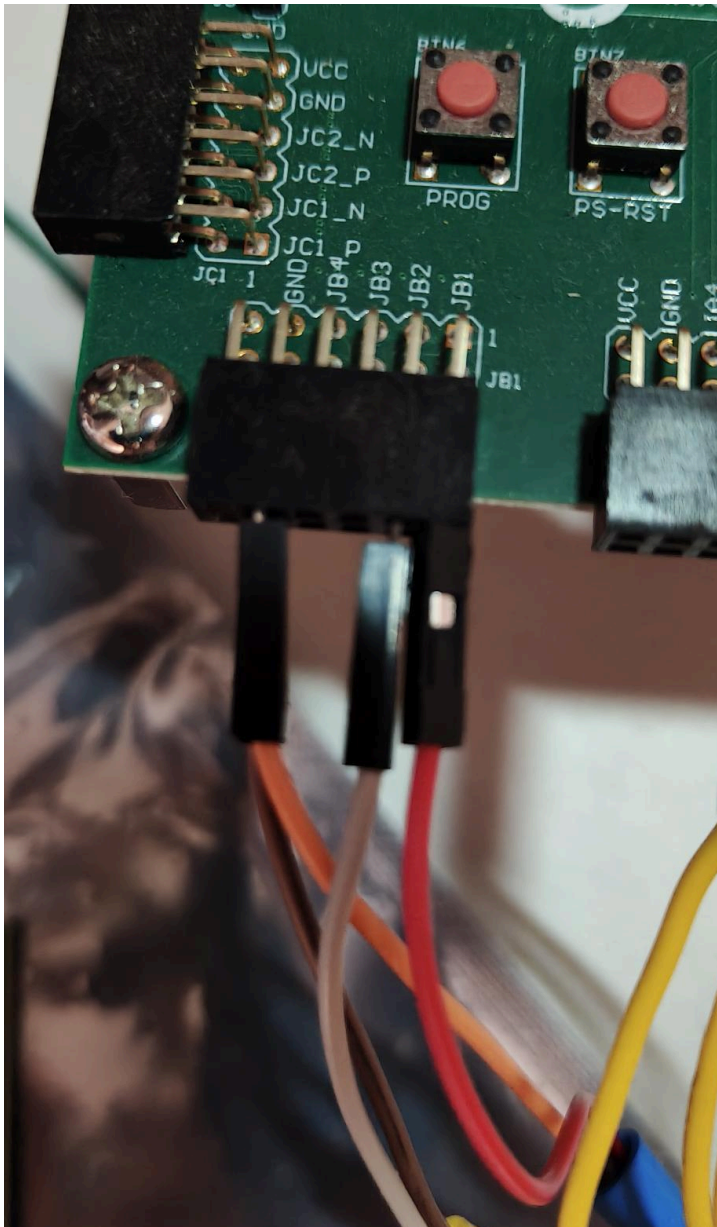
Getting started guide for the zedboard setup.

The hardware setup

First, attach the FMC2ZMOD adapter board to the Zedboards FMC Port
Then, plug the TCSPC module into port A of the FMC2ZMOD adapter board.

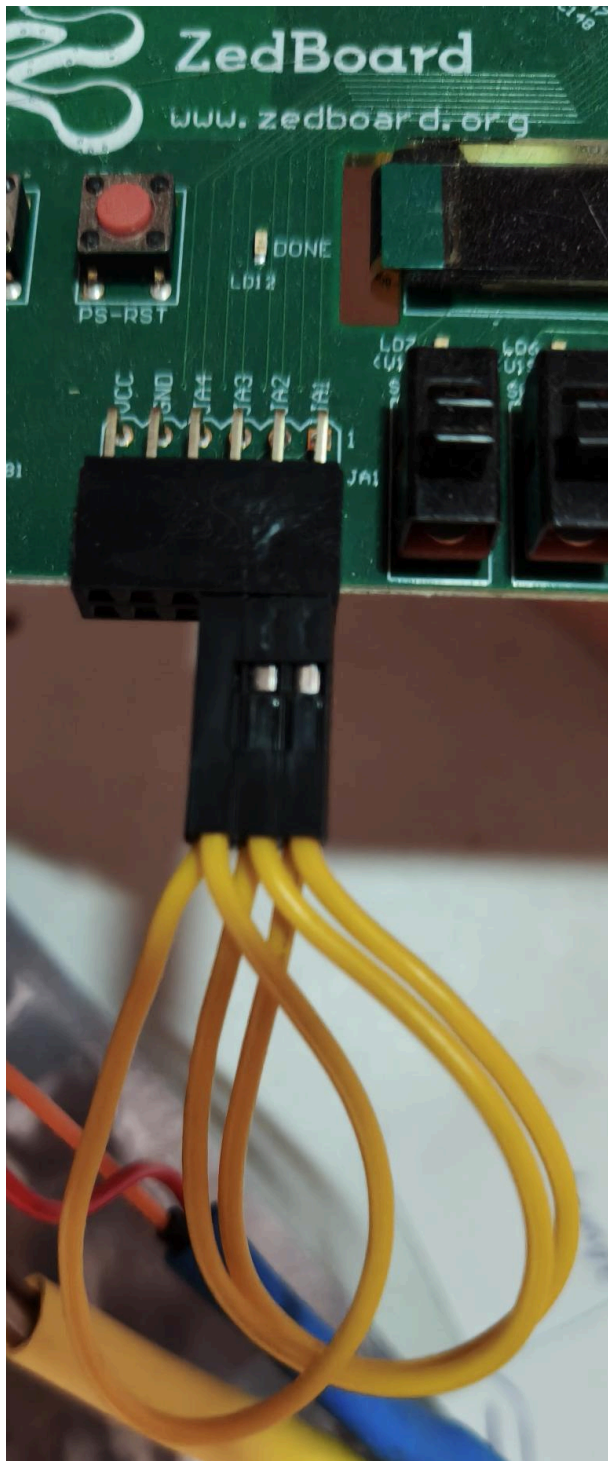


The Stop1 and Refclk inputs of the TCSPC Module (using SMB-Connectors) are to be connected to the microscopes detector and pulsed laser driver. Make sure not to go above 3.3V so that you do not fry the zedboard input pins. For using the microscope emulator on the FPGA, instead of a real microscope setup, these wires have to be connected to PMOD B of the Zedboard: the Refclk signal goes to JB1, while the Stop1 signal goes to JB2. The Ground lines of the wires can also be connected to the ground connections of PMOD B, namely JB5 and JB11.

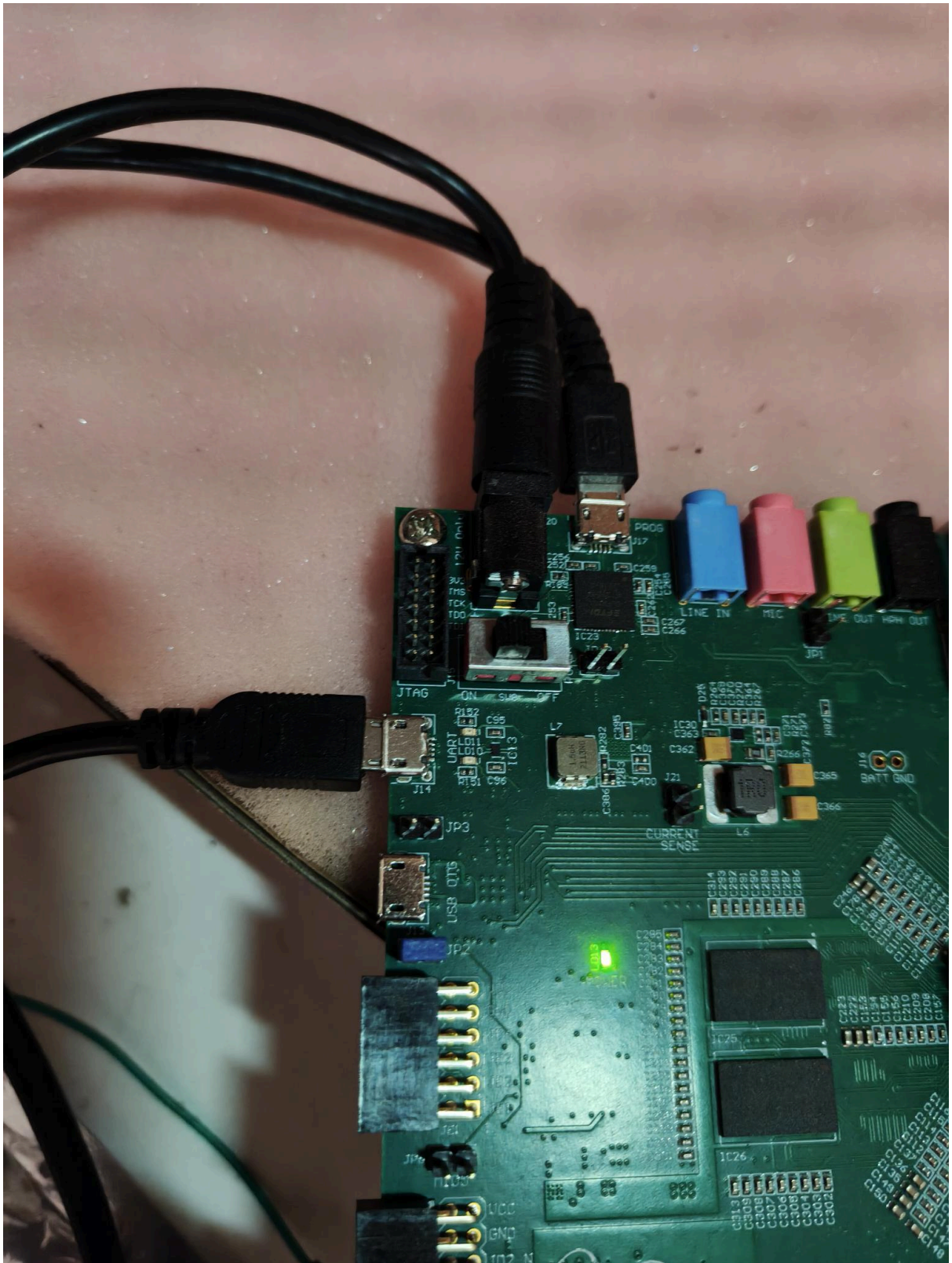


The inputs signals for frame_clk, line_clk, pixel_clk from the microscopes scanning optics go to PMOD A's JA1, JA2, JA3, respectively.

when using the microscope emulator, connect the other end of these signals to JA7, JA8, JA9, respectively.



finally, connect both zedboard usb cables for programming and serial port, to the pc, as well as the 12V power supply and turn the zedboards main power switch on



FPGA code

either just work with the generated .xla file in
openTCSPC\SRC\zedboard_real_microscope\vitis

or open the main vivado project zedboard_real_microscope.xpr in
openTCSPC\Projects\zedboard_real_microscope

if this project file does not exist yet, you have to build it using hdl on git (hog).

using a linux terminal (or git bash for windows) type:

```
./Hog/Do CREATE zedboard_real_microscope
```

For instructions how to setup hog, see:

<https://hog.readthedocs.io/en/latest/01-Getting-Started/01-howto-existingProjects.html>

the vivado project uses 3 IP modules which are also included in this repository:

The Microscope Emulator IP, the LVDS_to_AXIS IP, and the Histogramming IP.

these can be found in openTCSPC\Projects\ or be built using hog with the command:

```
./Hog/Do CREATE Microscope  
./Hog/Do CREATE histogramming  
./Hog/Do CREATE LVDS_to_AXIS_IDDR
```

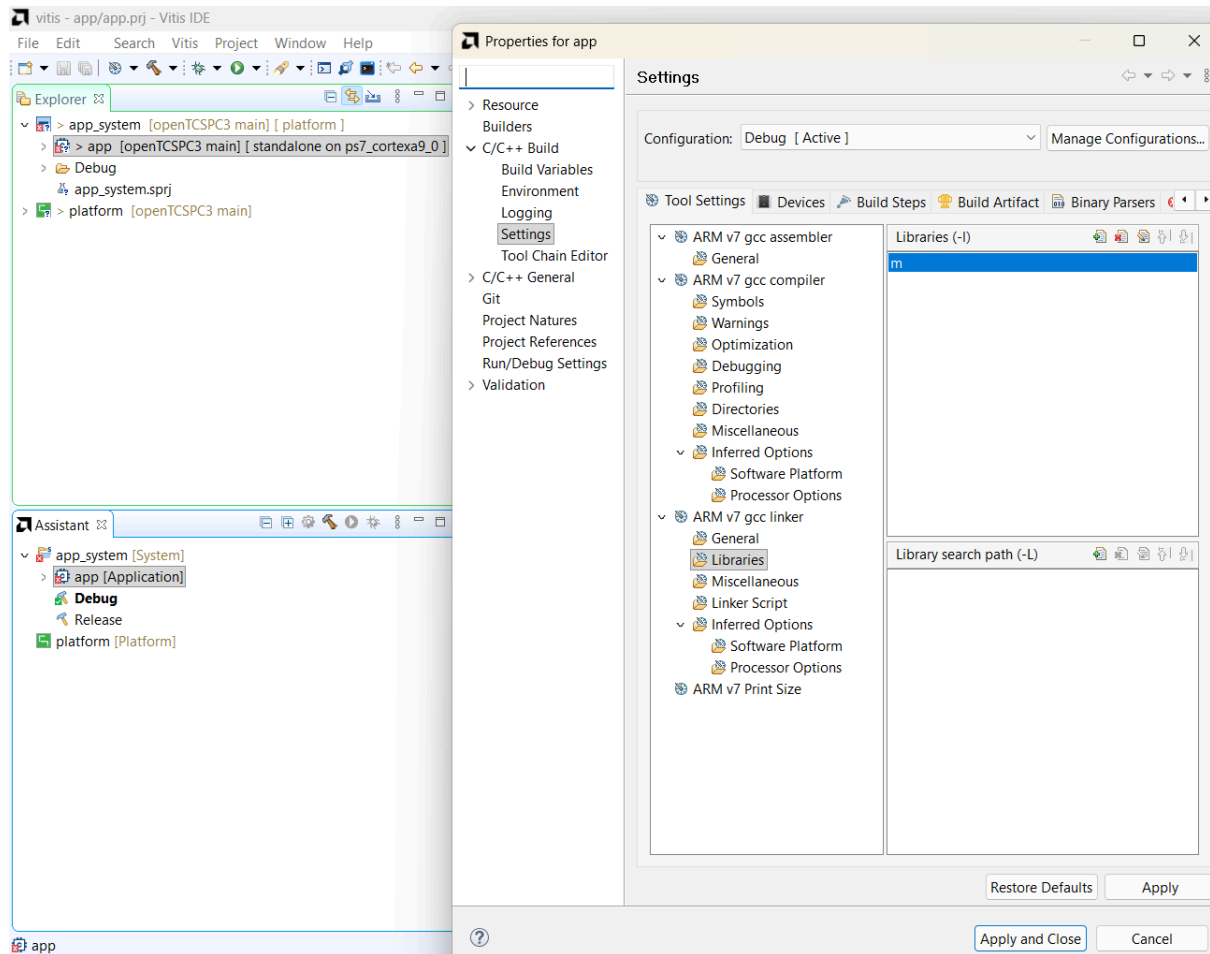
However, the files in the openTCSPC/IP folder can be used as is.

C code

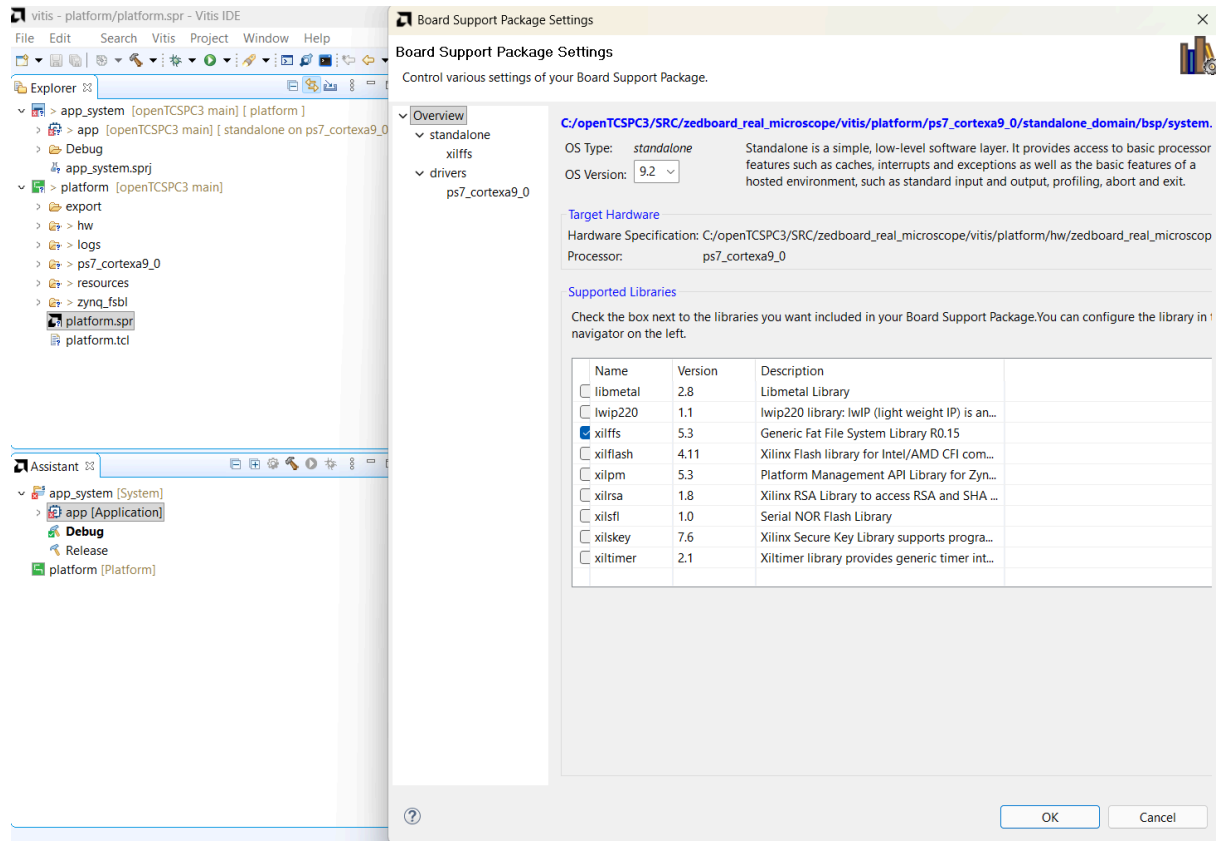
after exporting (or just using the provided) .xla file, the vitis project can be opened in vitis classic 2024.2 the project, along with all its source code is
openTCSPC\SRC\vitis\zedboard_real_microscope

when building a new vitis project, two things have to be paid attention to:

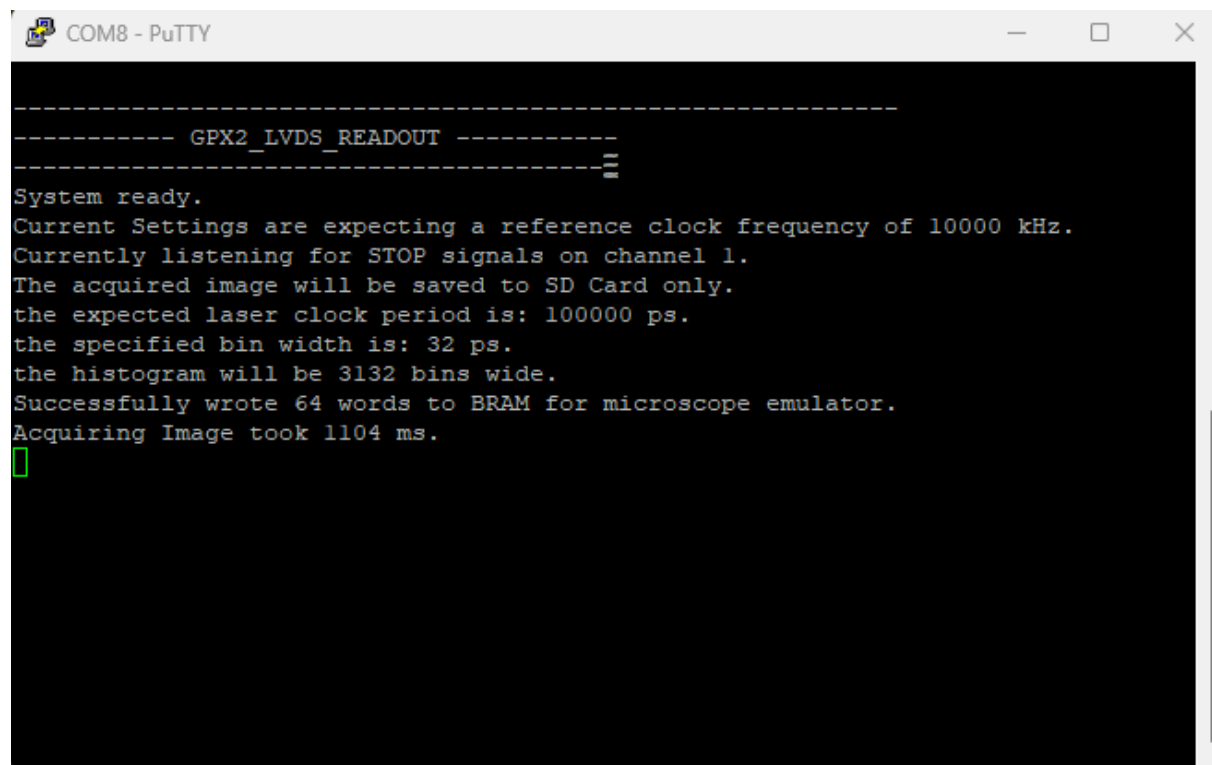
1: you need to include the math.h library in the application projects C/C++ build settings:



2: you need to include the xilffs file system in the bsp settings:



run the program on the zedboard and monitor the serial output(115200 baud):



in the C code, the `global_defines.h` file is where most of the important parameters can be set:

laser frequency, image size, time-to-digital converter resolution, histogram bin width etc...

Python Code

to visualize and evaluate the acquired image, the `image_analysis.ipynb` jupyter notebook is provided in the `python` subdirectory.